

TRAINING-FREE · LOCAL REFINEMENT · FROZEN GENERATOR

Small regions. Finer detail.

Refine the geometry and texture that matter,
while preserving the surrounding 3D asset.

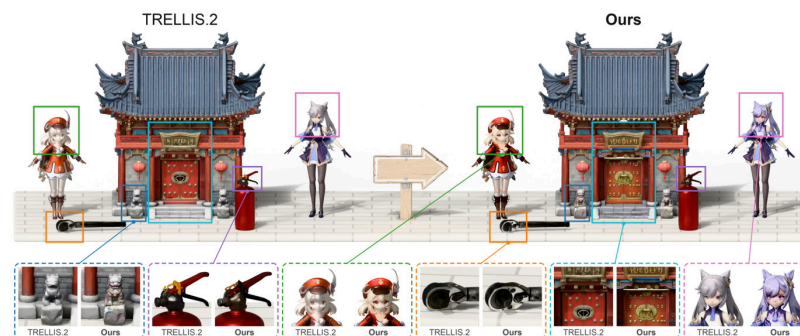
NOFACE3D: Noise-Scheduled Focused Anchoring and Controlled Enhancement for 3D Generation

[Authors and affiliations / anonymous review version]

[Read the paper ↗](#)[Explore results ↓](#)[Code · planned release](#)[BIGDETAIL](#)

A CLOSER LOOK AT GENERATED 3D

PAPER FIGURE 1 · STATIC PREVIEW



Given a generated mesh, its conditioning image, and a selected region, NOFACE3D reallocates the generator's spatial capacity to recover finer local detail.

01 / RESULTS **Get closer. See the difference.**

Explore improvements in both appearance and geometry, from character details to compact mechanical structures.

[Objects · BIGDETAIL](#)[Characters & general assets](#)

BIGDETAIL comparisons

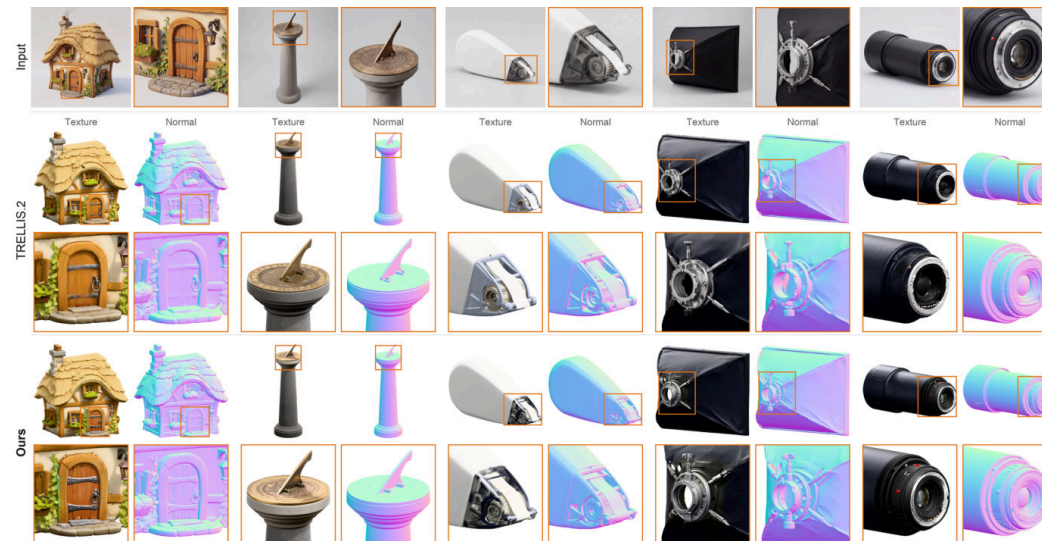
[PAPER PREVIEW](#)

Figure 4 · Input regions, original results, and refinements in textured and normal views.

Interactive comparison — reserved space

Replace the paper preview above with paired 3D assets. Controls below show the intended interface and are not connected.

Original · TRELLIS.2

[Original GLB + matching camera]

Refined · NOFACE3D

[Refined GLB + matching camera]

[Texture](#)[Clay](#)[Normals](#)[Full asset](#)[Detail](#)[Rotate together](#)

02 / METHOD More capacity, where it matters.

A small region occupies the generator's full input volume. Boundary anchoring keeps regeneration consistent with the surrounding asset.

01

Select a region

Mark a detail-critical region in the mesh and conditioning image.

03

Refine with anchoring

Hold boundary latents close to the base; allow stronger changes farther away.

02

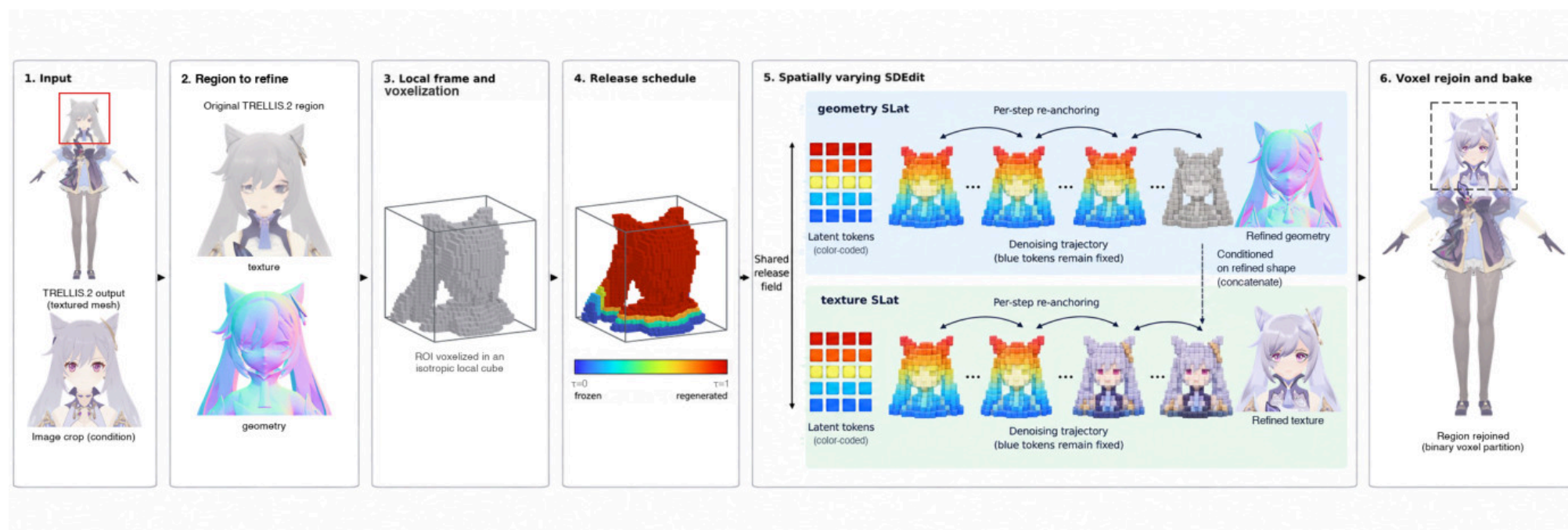
Enlarge & re-encode

Let the selected submesh fill the generator's existing spatial grid.

04

Reintegrate

Return the refined region to the original coordinate frame and combine it with the asset.



03 / ABLATION

The boundary makes the difference.

Regenerating a crop alone can leave visible discontinuities. Spatial noise scheduling encourages a more coherent transition.



Figure 5 · Without scheduling (top), with scheduling (bottom). Matched detail crops reveal the attachment region.

04 / EDITING A new detail. Then another.

An edited image crop turns the same refinement pipeline into a local 3D editor. Successive edits can target different regions.

Hat → robotic legs

Wings → head refinement

Engine bay → rockets

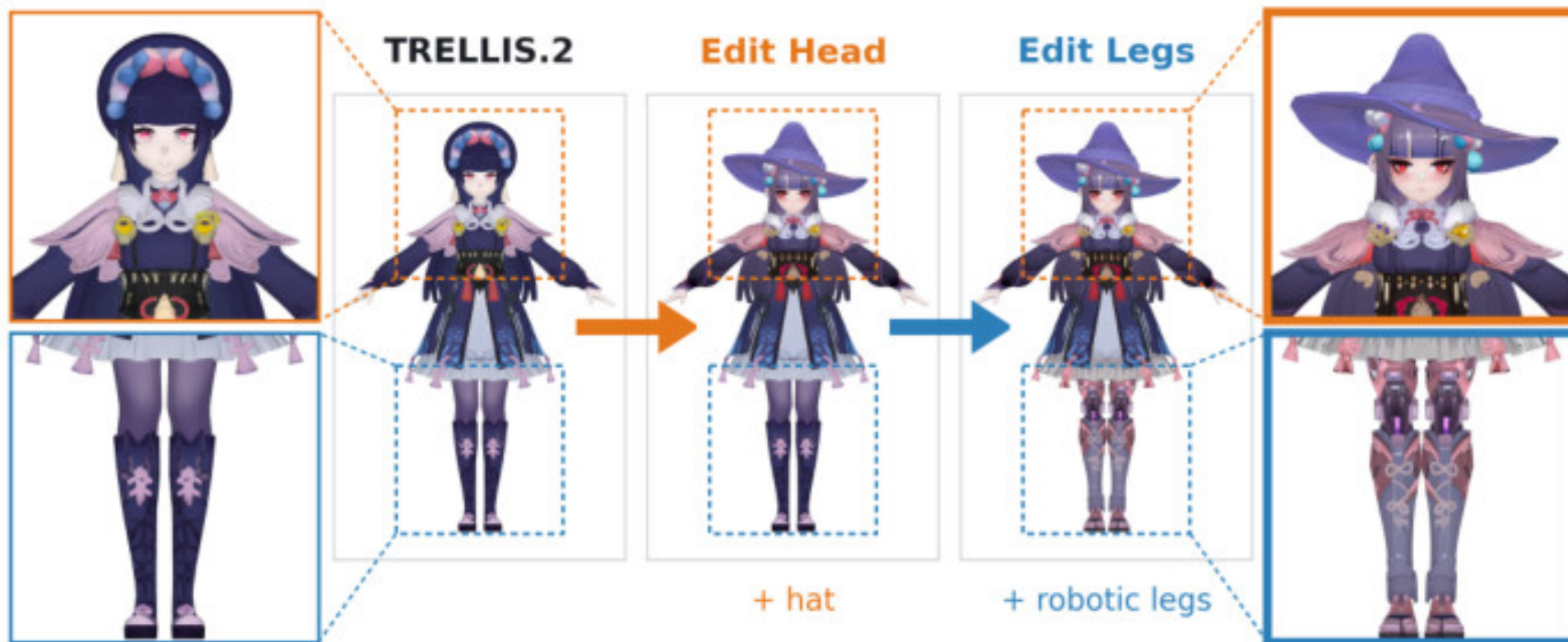


Figure 6 · The second edit targets the legs while retaining the earlier hat edit. Conditioning crops are edited with Gemini Flash Image.

05 / EVALUATION

Visible detail. Measured preference.

83.3%

Human preference

Of judged ratings preferred NOFACE3D. Including “neither,” 73.3% of all responses preferred our method.

4,892

PAniC-3D assets

Large-scale evaluation against the original TRELIS.2 generations.

339

BIGDETAIL assets

Refined assets with annotated detail-critical regions in images and meshes.

► Study context and selected quantitative results

06 / BENCHMARK

Meet BIGDETAIL.

339 REFINED ASSETS

Small parts. A broad set of objects.

A benchmark built around large, smooth forms with compact, detail-rich regions: mechanisms, fittings, connectors, and more.

Conditioning images

Generated object images and corresponding detail crops.

[Dataset · planned release](#)[Preview examples ↑](#)

Region annotations

Image rectangles and oriented 3D selection boxes.

Evaluation protocol

Paired comparisons of original and refined assets; no artist ground-truth meshes.

Where refinement still struggles

The method inherits the frozen generator's limitations. Printed labels and lettering may remain distorted after refinement.

Read, reproduce, build on it.

- Paper ↗
- Code · planned release
- Dataset · planned release

► Citation — metadata pending

NOFACE3D · Research project page

Template uses figures from the supplied draft · [Credits / contact]